

ShopEZ – Full Stack Development Documentation

Introduction

ShopEZ is a full-featured e-commerce platform developed using the MERN stack. The platform provides customers with an intuitive shopping experience while offering administrative tools for managing products, users, and orders.

System Features

- User Registration & Login
- Product Browsing
- Advanced Search
- Shopping Cart
- Checkout
- Order History
- Admin Dashboard
- Analytics

System Architecture

- Frontend: React.js
- Backend: Express.js & Node.js
- Database: MongoDB
- Authentication: JWT
- Communication: REST APIs using Axios

Database Design

Collections include Users, Products, Orders, Reviews, Categories and Cart. Indexes are applied to product names and user emails for performance.

Security

JWT Authentication, Password Hashing with bcrypt, Protected Routes, Input Validation, Role-Based Access Control and CORS Protection.

Testing Strategy

Unit Testing, Integration Testing, User Acceptance Testing, Responsive Testing, Performance Testing and Security Validation.

Future Enhancements

AI Recommendations, Payment Gateway Integration, Real-Time Notifications, Mobile App, Multi-Vendor Marketplace and Progressive Web App support.